

STORYALIGN: EVALUATING AND TRAINING REWARD MODELS FOR STORY GENERATION

Haotian Xia*, Hao Peng*, Yunjia Qi, Xiaozhi Wang, Bin Xu[†], Lei Hou, Juanzi Li

Department of Computer Science and Technology, Tsinghua University

{xiaht24, peng-h24}@mails.tsinghua.edu.cn

{xubin, lijuanzi, houlei}@tsinghua.edu.cn

ABSTRACT

Story generation aims to automatically produce coherent, structured, and engaging narratives. Although large language models (LLMs) have significantly advanced text generation, stories generated by LLMs still diverge from human-authored works regarding complex narrative structure and human-aligned preferences. A key reason is the absence of effective modeling of human story preferences, which are inherently subjective and under-explored. In this work, we systematically evaluate the modeling of human story preferences and introduce STORYRMB, the first benchmark for assessing reward models on story preferences. STORYRMB contains 1,133 high-quality, human-verified instances, each consisting of a prompt, one chosen story, and three rejected stories. We find existing reward models struggle to select human-preferred stories, with the best model achieving only 66.3% accuracy. To address this limitation, we construct roughly 100,000 high-quality story preference pairs across diverse domains and develop STORYREWARD, an advanced reward model for story preference trained on this dataset. STORYREWARD achieves state-of-the-art (SoTA) performance on STORYRMB, outperforming much larger models. We also adopt STORYREWARD in downstream test-time scaling applications for best-of-n (BoN) story selection and find that it generally chooses stories better aligned with human preferences. We will release our dataset, model, and code to facilitate future research. Related code and data are available at <https://github.com/THU-KEG/StoryAlign>.

1 INTRODUCTION

The task of story generation aims to automatically generate coherent, structured, and engaging narratives from a given premise (Wang et al., 2023). With the rise of large language models (LLMs) and their powerful text generation capabilities, many studies leverage them and design workflows to generate fluent stories (Alhussain & Azmi, 2021; Wang et al., 2024b; Lee et al., 2025; Xia et al., 2025; Huot et al., 2025). However, LLM-generated stories are often considered to lack complex narrative structure and creativity (Chakrabarty et al., 2024; Tian et al., 2024; Wang & Kreminski, 2025). For example, Chakrabarty et al. (2024) report that LLM-generated stories are three to ten times less likely to pass expert evaluation compared with those written by professional authors. This gap highlights that, despite significant progress, LLM-generated narratives remain far from human standards and preferences.

The main reason is that current LLMs lack effective modeling of human story preferences during training. LLMs typically rely on reinforcement learning from human feedback (RLHF; Ouyang et al., 2022), which first

* Equal contribution.

[†] Corresponding author.

trains a reward model (RM) to capture human preferences and then optimizes LLMs based on these rewards. However, existing reward models mainly focus on general domains (Zhong et al., 2025) and may overlook story-specific preferences. Meanwhile, they also exhibit biases, such as verbosity bias (Saito et al.; Peng et al., 2025), which favors longer rather than higher-quality responses. As a result, models trained with such reward signals may inherently struggle to generate stories aligned with human preferences. Moreover, modeling story preferences is inherently challenging, as it is subjective and spans multiple dimensions such as creativity and relevance. The automated evaluation of LLM-generated stories thus remains an open problem (Yang & Jin, 2024). Therefore, effectively evaluating and improving story preference modeling of RMs is necessary.

To address this gap, we systematically evaluate the modeling of human story preferences and develop an advanced reward model for story generation. Specifically, we introduce STORYRMB, the first benchmark for evaluating reward models of story preferences. Inspired by prior work (Yang & Jin, 2024), we define 5 evaluation criteria for STORYRMB, including coherence, creativity, characterization, fluency, and relevance. The construction of STORYRMB consists of two parts: (1) Candidate story generation, which collects diverse stories for the same premise from 4 advanced LLMs, including Gemini 2.5 Pro (Comanici, 2025), Grok 3 (xAI, 2025), GPT-4o (OpenAI, 2024), Qwen-Long (Wan et al., 2025a). We also include human-written stories for about 250 premises as candidates to ensure real-world coverage. (2) Preference judgment, where we design a two-stage semi-automated annotation process. We use majority voting among four advanced LLMs, including GLM-4.5 (Team et al., 2025), Qwen2.5-Max (Yang et al., 2025), Claude-3.5-Sonnet (Claude, 2024), and Deepseek-R1 (DeepSeek-AI et al., 2025), to select a chosen story, followed by human verification. For cases of strong disagreement, the preferences are directly annotated by humans. In total, STORYRMB comprises 1,133 high-quality and human-verified instances, each containing a premise, one chosen story, and three rejected stories. We then evaluate several advanced reward models on STORYRMB. We find that existing reward models struggle to reliably select human-preferred stories, with the highest accuracy at only 66.3%. We also find that existing reward models tend to favor LLM-generated over human-written stories, making it difficult to guide models toward human-level story generation (Tian et al., 2024).

Consequently, we propose STORYREWARD, an advanced reward model for story preferences, trained upon a large-scale, high-quality dataset of story preference pairs. To ensure real-world coverage and accurately capture human preferences, we collect a large number of human-written stories along with corresponding human preference signals. Specifically, we collect stories and associated metadata, such as category and upvote counts, from Chinese and English online literary platforms, and construct preference pairs using three methods: (1) Premise back-generation. We randomly sample story pairs from the same category with available upvote information and use an LLM to back-translate corresponding premises (Qi et al., 2024). The story with more upvotes is denoted as chosen, and the other as rejected, yielding about 6,000 preference pairs. (2) Prompt-guided rewriting. We randomly select a premise and its corresponding human-written story as chosen, then adopt an LLM to rewrite the story as rejected. We ensure the rewriting is constrained via prompts to limit modifications, e.g., to the ending or title, preventing trivial discrepancies between chosen and rejected stories. This approach assumes human-written stories are generally preferred, which we also validate on 20 samples. This results in about 30,000 preference pairs. (3) Human-guided continuation. We sample human-written stories and use their first part as a premise for continuation. We adopt the original human-written continuation to guide the LLM’s output, which is treated as chosen, while the LLM’s continuation generated directly from the premise is treated as rejected. We do not use the human-written continuation directly as chosen to avoid trivial discrepancies, as its quality is significantly higher than the LLM’s continuation. This yields about 10,000 preference pairs. We also adopt a widely-used preference-pair construction method (Cui et al., 2024). We sample two stories for a given premise and adopt an LLM to select the best as chosen. In total, we obtain 100,000 high-quality preference pairs capturing diverse, real-world human preferences, which are then used to train STORYREWARD. STORYREWARD achieves state-of-the-art performance on STORYRMB, even surpassing much larger models. We further apply STORYREWARD in test-time scaling, conducting Best-of-N (BoN) experiments where reward models select a better story. Experimental results show that stories chosen by

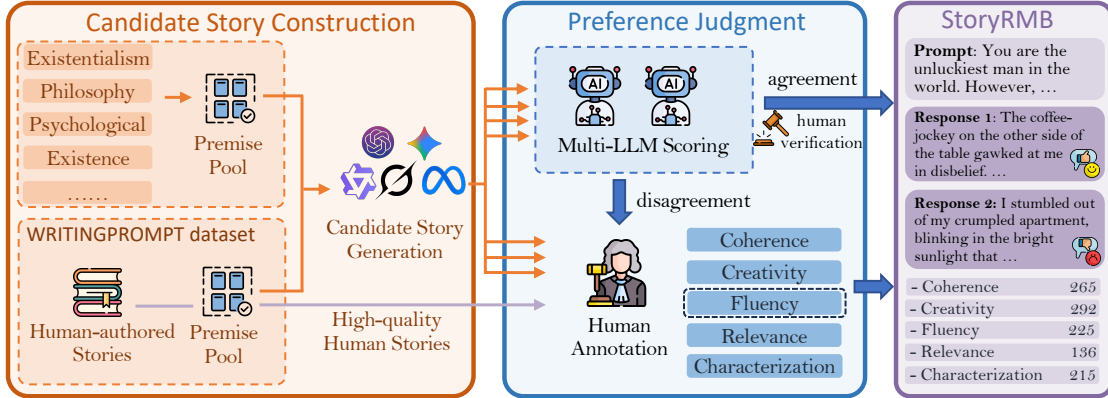


Figure 1: An overview of the benchmark construction framework. The process consists of: (1) the collection of candidate stories generated by LLMs and humans; (2) scoring the stories and partitioning them along various dimensions. These two stages yield a diverse dataset for evaluating the story reward model.

STORYREWARD substantially outperform those from other reward models, further validating its effectiveness in real-world applications. In conclusion, our contributions are mainly threefold:

- (1) We introduce STORYRMB, the first benchmark for assessing reward models on modeling story preferences. Extensive experiments demonstrate that existing reward models struggle to capture human story preferences.
- (2) We propose an automated method for collecting story preference pairs and build a large-scale, high-quality training dataset that captures diverse, real-world human story preferences.
- (3) We propose STORYREWARD, an advanced reward model for story preference that achieves SoTA performance on STORYRMB. We further validate its effectiveness in real-world applications.

2 STORYRMB: A BENCHMARK FOR STORY PREFERENCE

In this section, as illustrated in Figure 1, we present the methodology for constructing STORYRMB, including candidate story generation (§ 2.1) and a two-stage preference judgment process (§ 2.2).

2.1 CANDIDATE STORY CONSTRUCTION

Prompt Collection To generate diverse and high-quality stories, we first design a multi-stage **premise**¹ construction framework inspired by the modular synthesis approach of Ma et al. (2024). Our method begins by establishing a thematic framework drawn from literature and film, covering categories like Existentialism and Ethical Dilemmas. We then craft seed premises using a “conflict or reversal” structure (Fan et al., 2019; Barber & Kudenko, 2008) to be information-dense. These seeds are enriched with characters and settings, stylistically varied via LLM-based paraphrasing, and manually filtered for logical and cultural soundness. This process yields 624 high-quality premises. For more details, please refer to Appendix A.

In addition to the premises generated by LLMs, we add premises with human-written stories to enhance diversity. We randomly sample and filter 600 high-quality premises from the WritingPrompts dataset (Fan et al., 2018). Each of these premises is accompanied by a corresponding story authored by a human writer.

¹Premise refers to the story prompt, a term widely used in the story generation literature.